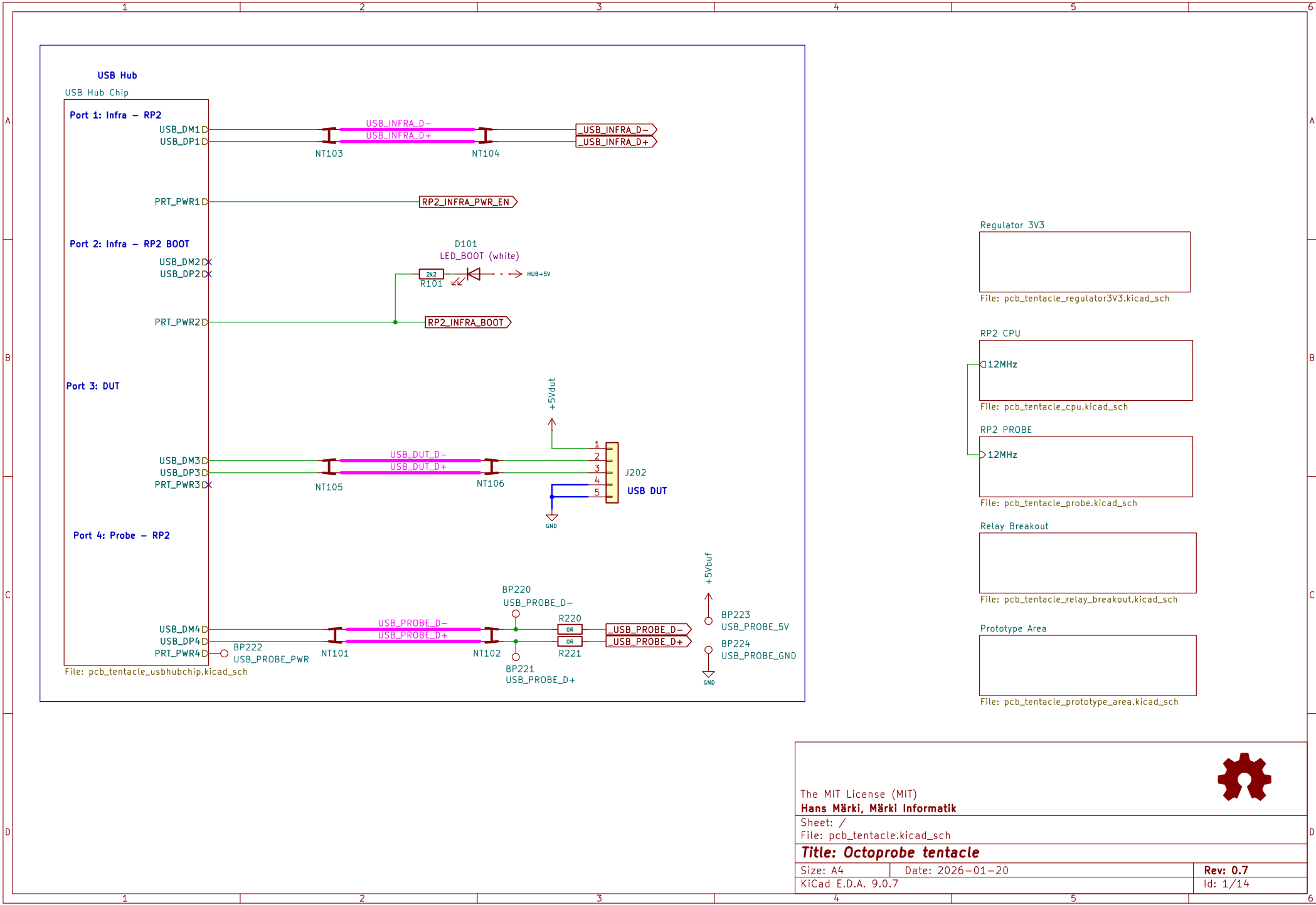
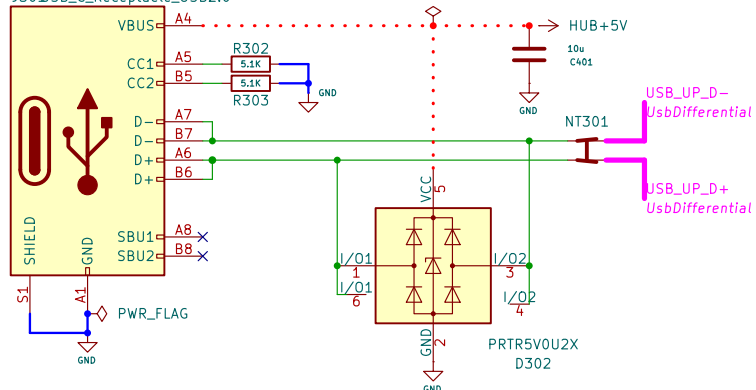




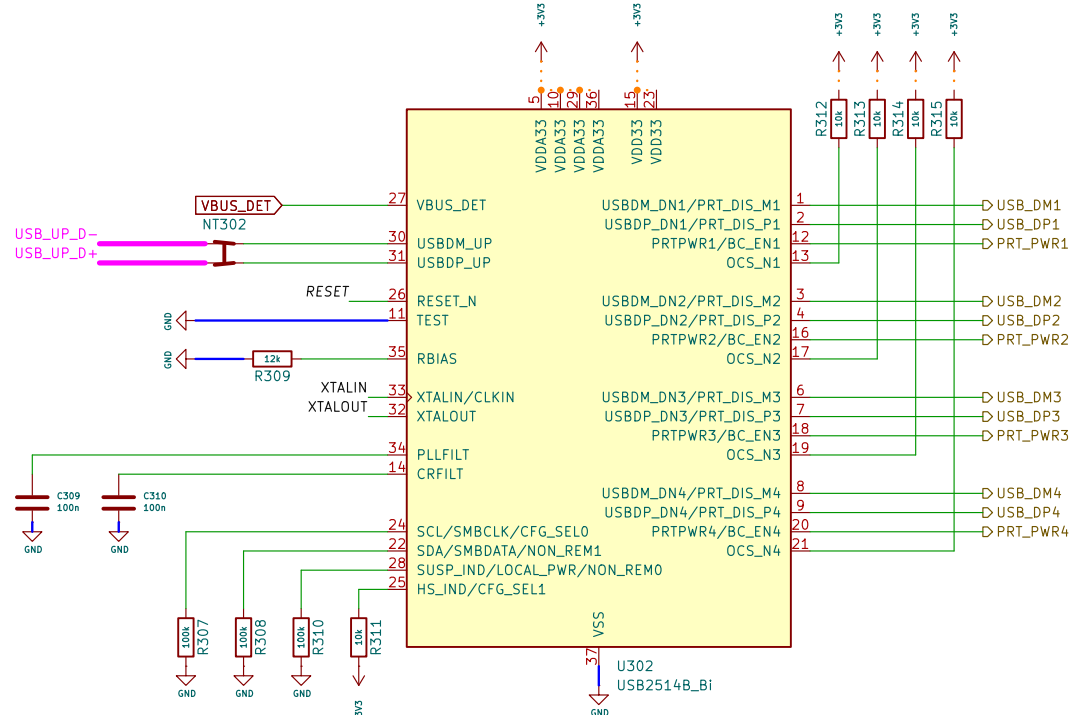
USB C Female

USB Type-C DATA

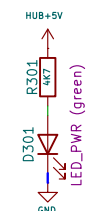
J301/USB_C_Receptacle_USB2.0



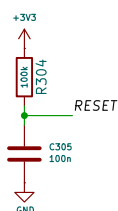
Hub



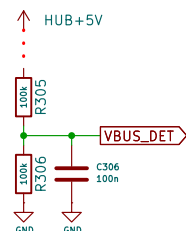
Power LED



Boot Sequence



VBUS_DET



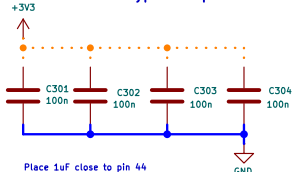
Scope GND Pad



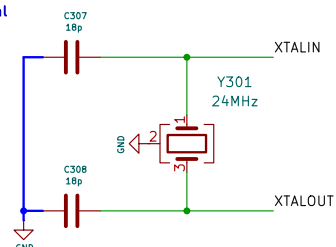
Microchip AN 15.17 "PCB Layout Guide for USB Hubs"

DP/DM:
27mil, 0.7mm, trace
5mil, 0.127mm, space
50mil, 1.27mm, trace side no ground

Bypass Capacitors



Crystal



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Hans Märki, Märki Informatik

Sheet: /USB Hub Chip/

File: pcb_tentacle_usbhubchip.kicad_sch

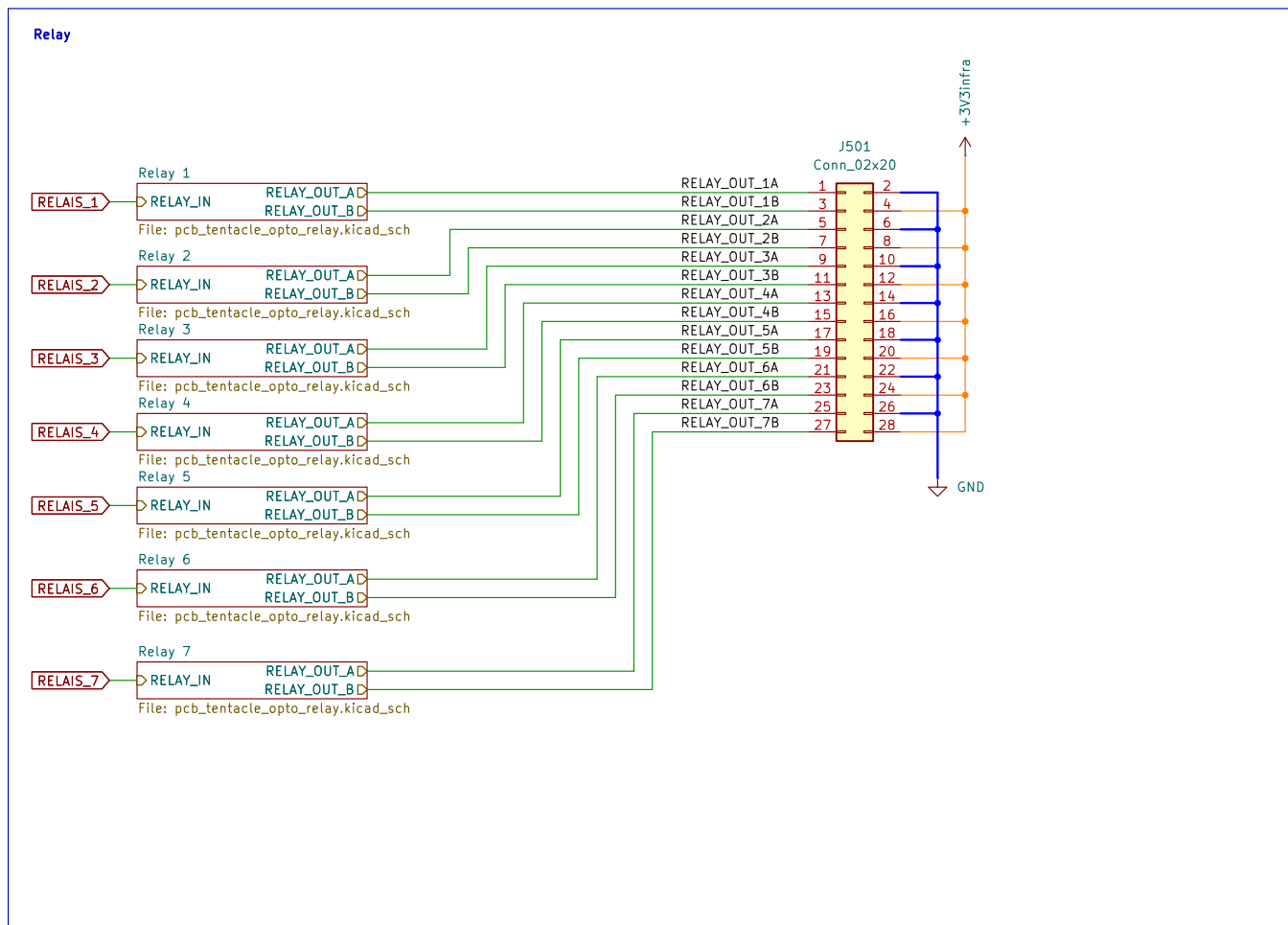
Title: Octoprobe tentacle

Size: A4 Date: 2026-01-20

KiCad E.D.A. 9.0.7

Rev: 0.7

Id: 3/14



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Hans Märki, Märki Informatik

Sheet: /Relay Breakout/

File: pcb_tentacle_relay_breakout.kicad_sch

Title: Octoprobe tentacle

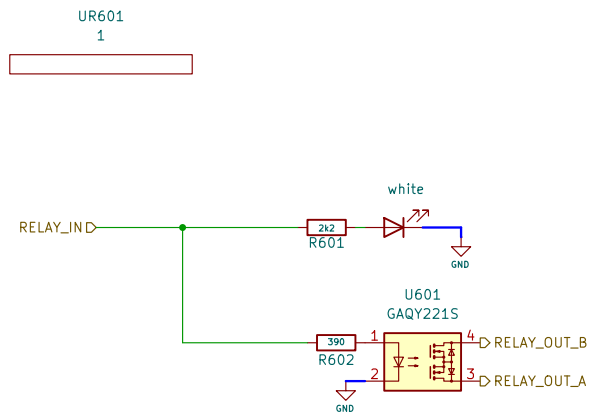
Size: A4

Date: 2026-01-20

Rev: 0.7

KiCad E.D.A. 9.0.7

Id: 5/14



Calculation of the input resistor

$V_{diff} = 3.3V - V_f = 3.3V - 1.2V = 2.1V$

$I_{Fon} = 5mA$

$R = 2.1V / 5mA = 420 \text{ Ohm}$

390 Ohm



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Sheet: /Relay Breakout/Relay 1/

File: pcb_tentacle_opto_relay.kicad_sch

Title: Octoprobe tentacle

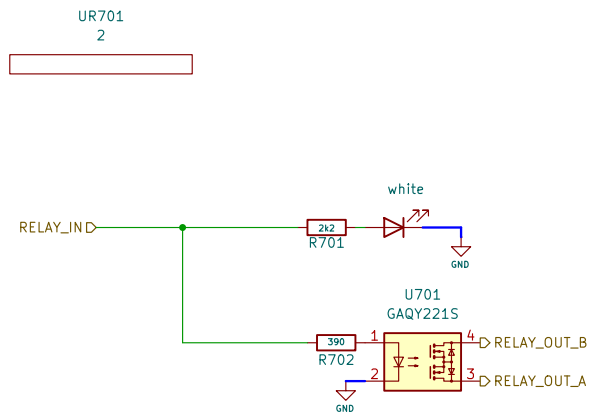
Size: A4

Date: 2026-01-20

Rev: 0.7

KiCad E.D.A. 9.0.7

Id: 6/14



Calculation of the input resistor

$V_{diff} = 3.3V - V_f = 3.3V - 1.2V = 2.1V$

$I_{Fon} = 5mA$

$R = 2.1V / 5mA = 420\ \Omega$

390 Ohm



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Sheet: /Relay Breakout/Relay 2/

File: pcb_tentacle_opto_relay.kicad_sch

Title: Octoprobe tentacle

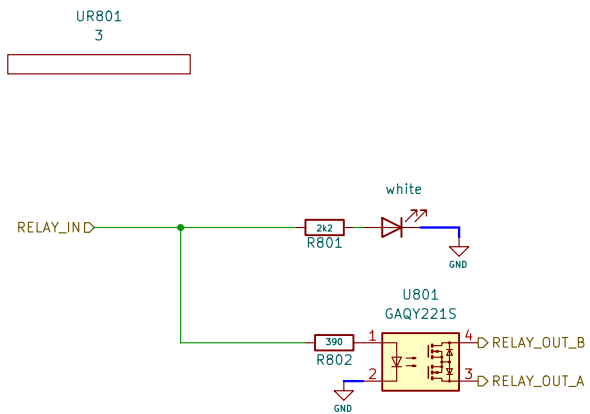
Size: A4

Date: 2026-01-20

Rev: 0.7

KiCad E.D.A. 9.0.7

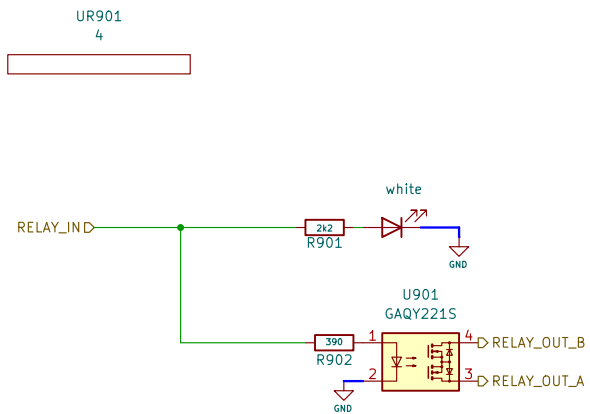
Id: 7/14



Calculation of the input resistor

$V_{diff} = 3.3V - V_f = 3.3V - 1.2V = 2.1V$
 $I_{Fon} = 5mA$
 $R = 2.1V / 5mA = 420\ \Omega$
 390 Ohm

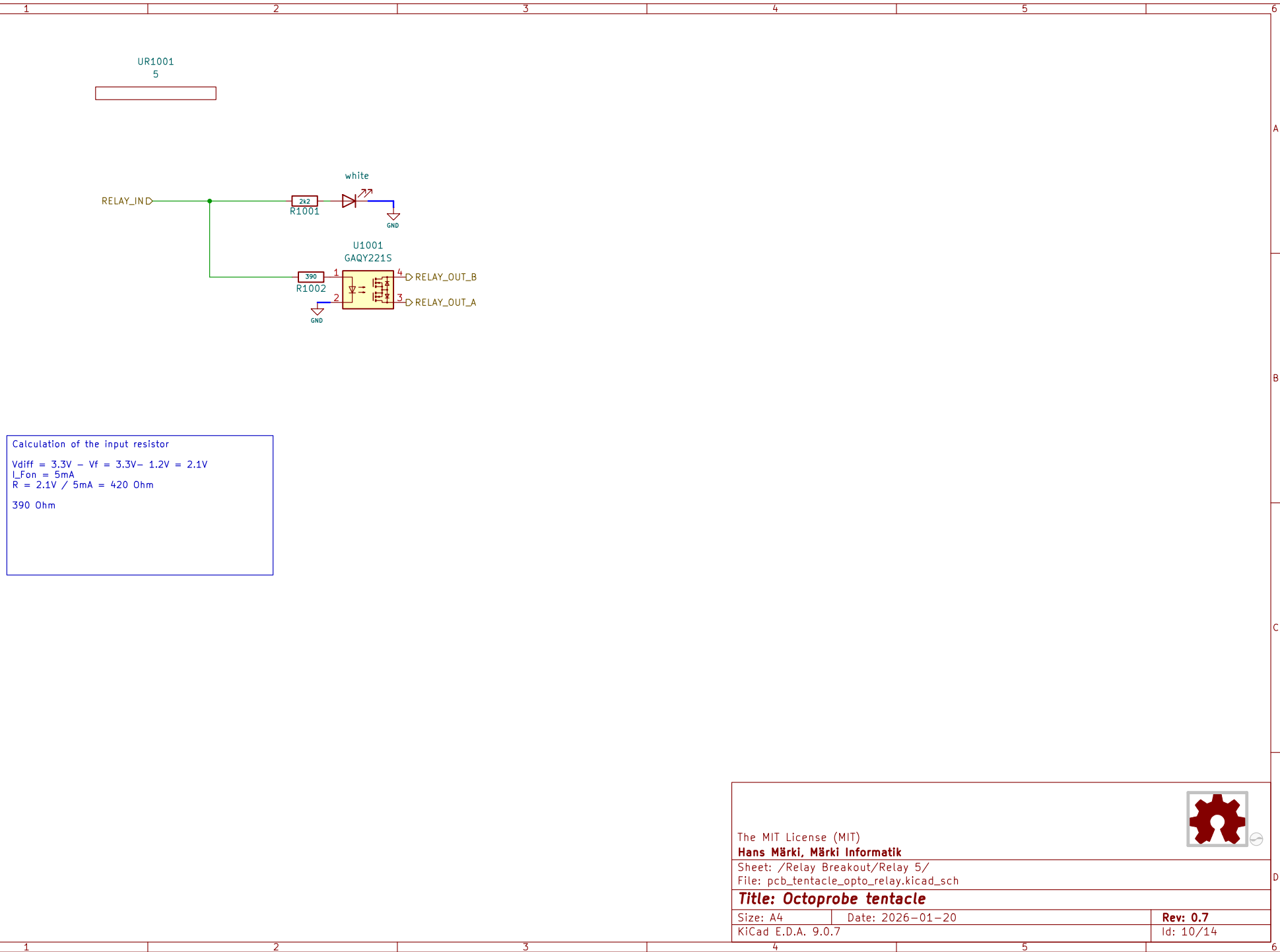


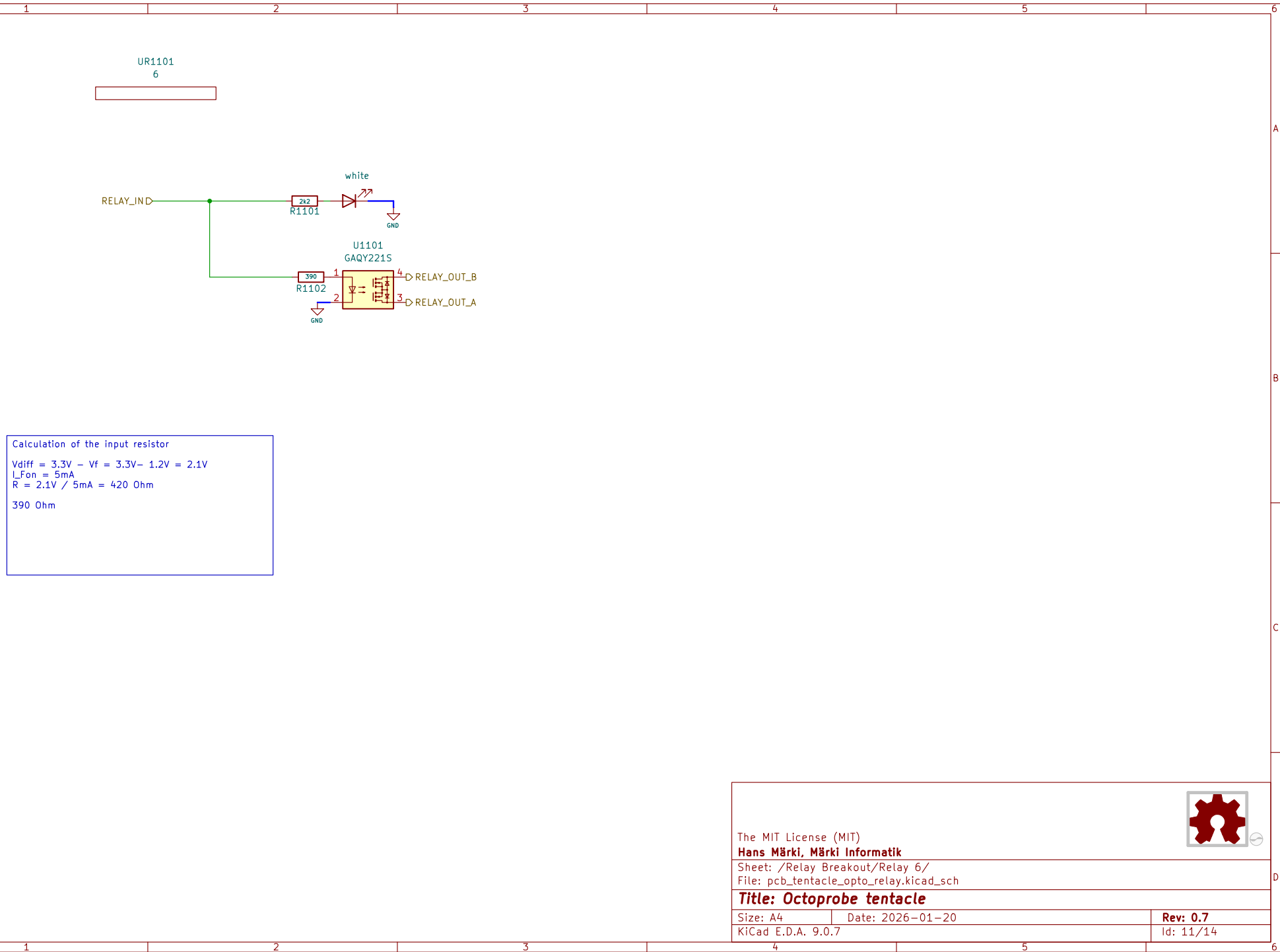


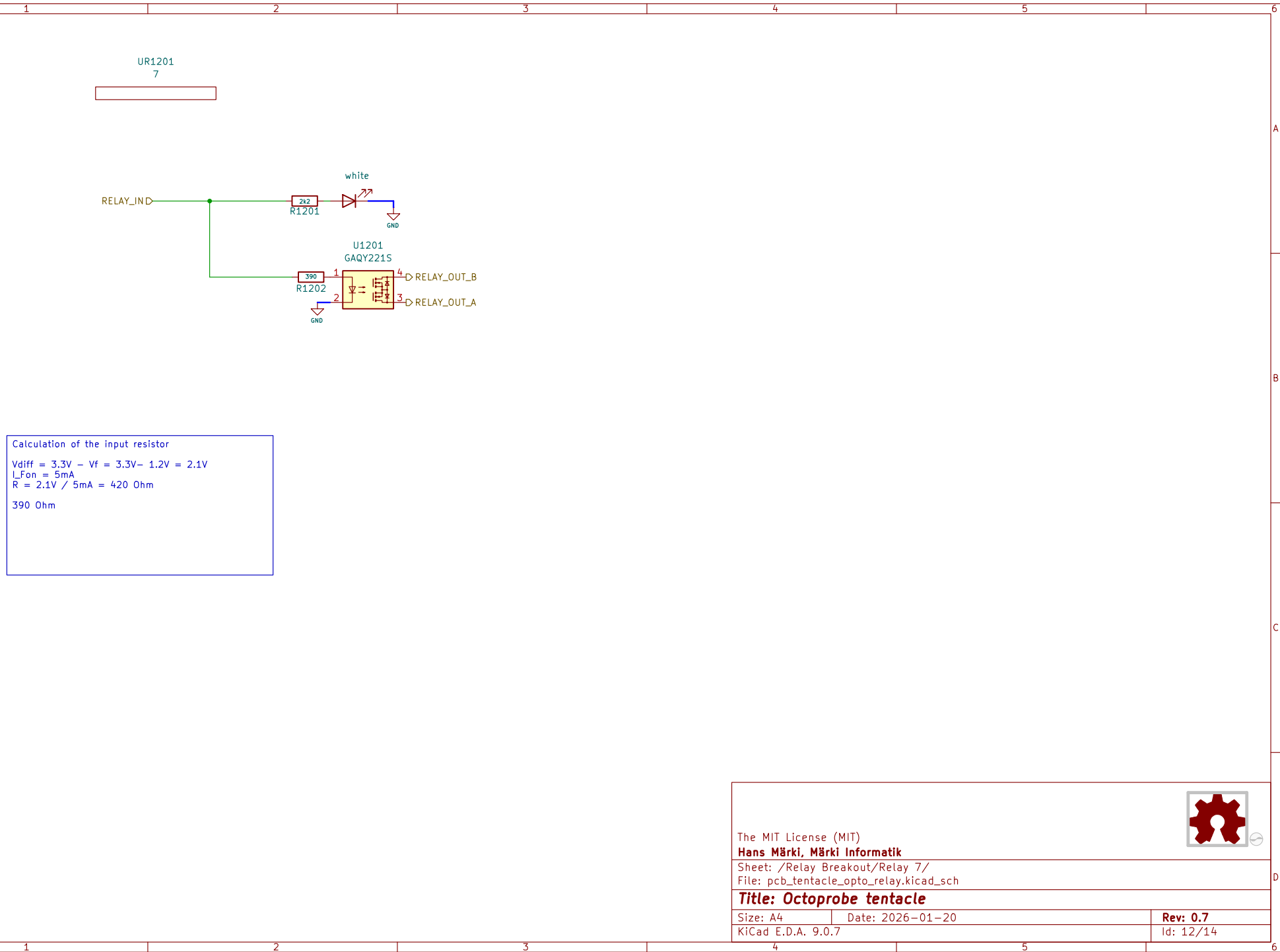
Calculation of the input resistor

$V_{diff} = 3.3V - V_f = 3.3V - 1.2V = 2.1V$
 $I_{Fon} = 5mA$
 $R = 2.1V / 5mA = 420\ \Omega$
 390 Ohm



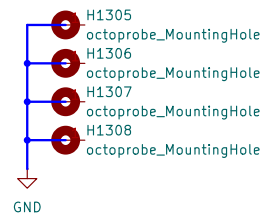






Prototype Area

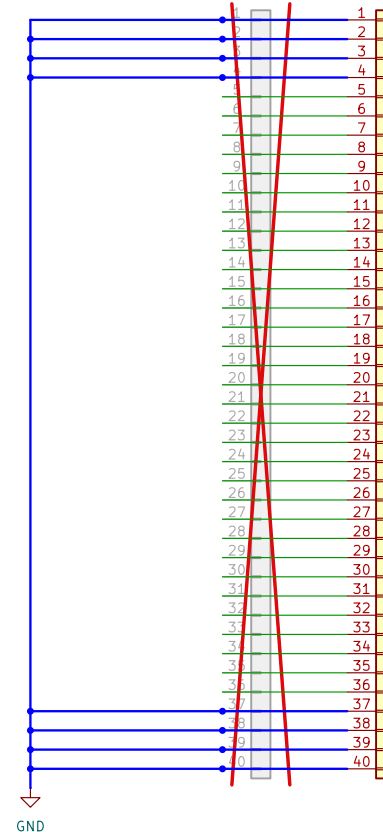
- TH1301
JLPCB Tooling Hole
- TH1302
JLPCB Tooling Hole



Octobus

J1302
Conn_02x20

J1303
Conn_02x20



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Sheet: /Prototype Area/

File: pcb_tentacle_prototype_area.kicad_sch

Title: Octoprobe tentacle

Size: A4

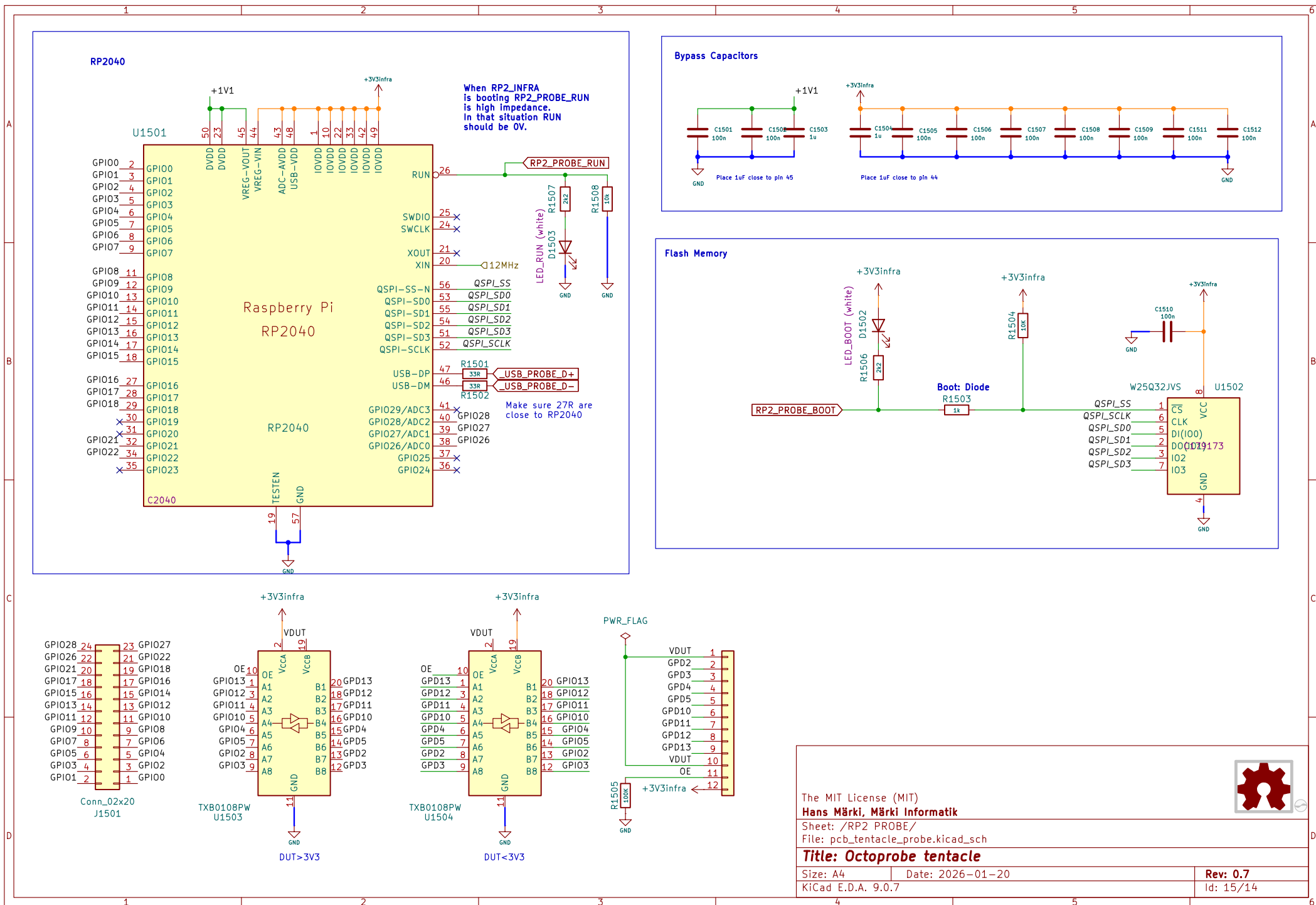
Date: 2026-01-20

Rev: 0.7

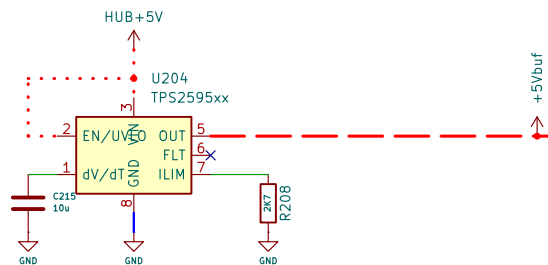
KiCad E.D.A. 9.0.7

Id: 13/14



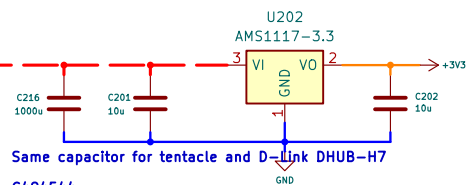


Current Limiter 5V



Datasheet "Active Current Limiting":
 Goal: 800mA
 $R_{lim} = 2000 / (0.8A - 0.04) = R2K630$, choosen: R2K7
 Datasheet "Slew Rate and Inrush Current Control":
 Goal: 5V/1000ms
 $C = 42000 / V * ms = 6'363'636pF = 6uF$, choosen: 10uF

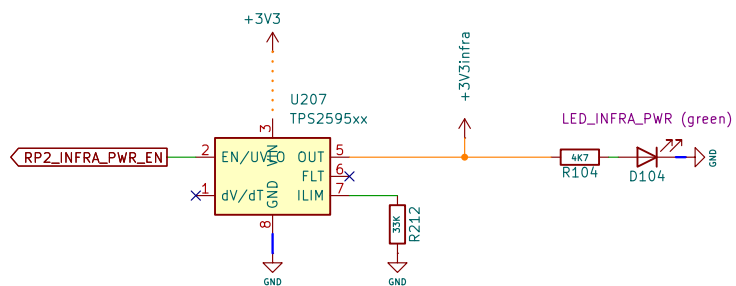
Regulator 3V3



Same capacitor for tentacle and D-Link DHUB-H7
 C494544

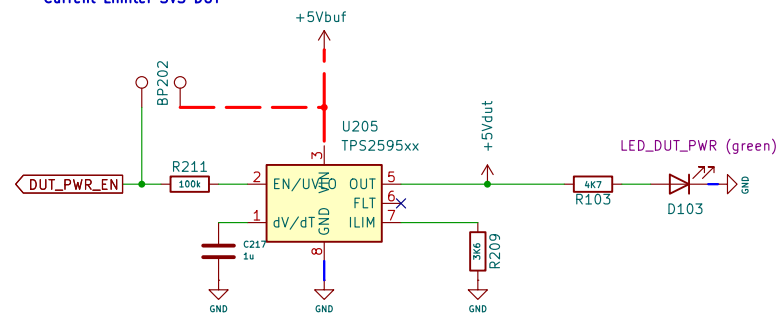
2.5mm distance between wires
 6.3mm diameter
 11mm height

POWER RP Infra



Datasheet "Active Current Limiting":
 Goal: 100mA
 $R_{lim} = 2000 / (0.1A - 0.04) = R33K333$, choosen: R33K

Current Limiter 3V3 DUT



Datasheet "Active Current Limiting":
 Goal: 600mA
 $R_{lim} = 2000 / (0.6A - 0.04) = R3K571$, choosen: R3K6
 Datasheet "Slew Rate and Inrush Current Control":
 Goal: 3V3/100ms
 $C = 42000 / V * ms = 1'272'272pF$, choosen: 1uF

Sheet: /Regulator 3V3/
 File: pcb_tentacle_regulator3V3.kicad_sch

Title:

Size: A4

Date:

Rev:

KiCad E.D.A. 9.0.7

Id: 16/14